

---

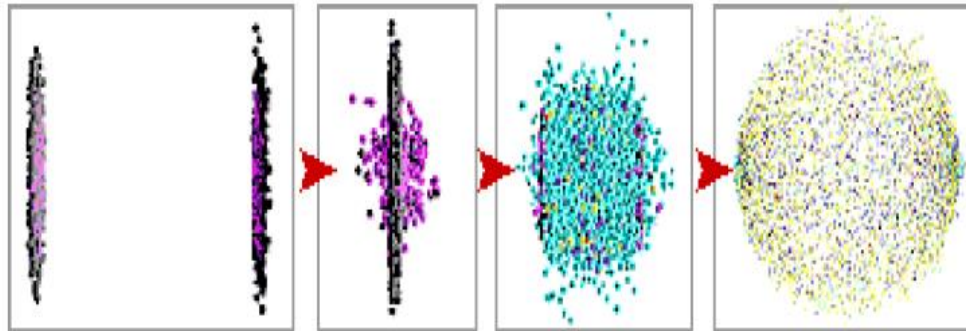
# Fluid mechanics in ATLAS

Laurent Schoeffel

---

## Chapter I

Collisions between 2 nuclei (PbPb scattering - *Pb*: 208 nucleons-)

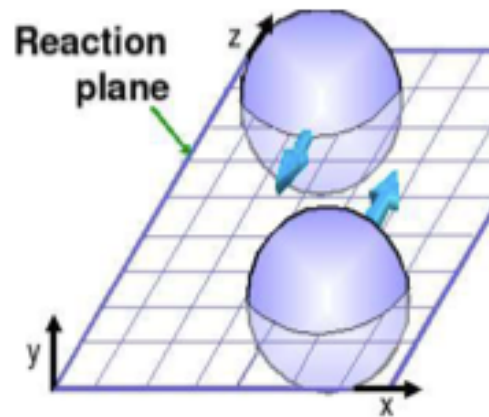


# Introduction

---

Let us assume that we have 2 heavy nuclei colliding as below. The beam axis is along (Oz).

The distance between the 2 centers of the nuclei along (Ox) is called the impact parameter  $\vec{b}$ .



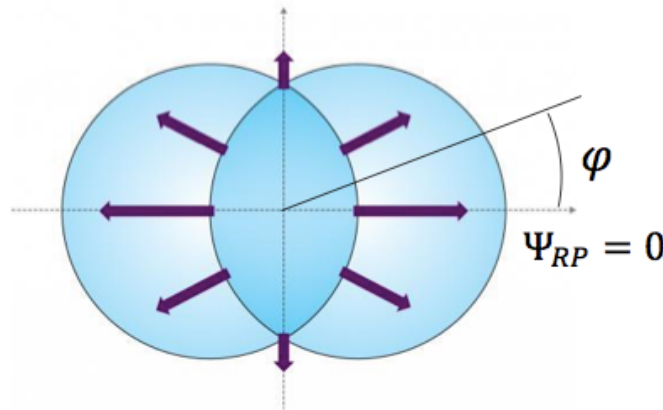
The plane  $(\vec{b}, \vec{e}_z)$  is called the reaction plane. We assume from now that there exists a reaction plane  $(\vec{b}, \vec{e}_z)$ .

This does not happen like this in the real world: we only assume this for the moment and will add more real inputs step by step...

## A first schematic view

The plane  $(\vec{b}, \vec{e}_z)$  is the reaction plane. This corresponds to  $\psi_{RP} = 0$  in the picture below.

Let us note that the modulus of  $\vec{b}$  can be more or less large: from 0 (central collision) to very large (greater than the size of the nuclei) for Ultra-Peripheral collisions. Here we take a medium value for illustration.



The key observation is that there is a shape in the middle (like an ellipsoid) where the constituents of the 2 nuclei enter in collision. This means that in this volume, a very large number of constituents are mixed... So, (i) what is the mixture in the collision volume and (ii) what can happen?

# The collision volume: first ideas

---

(1) Once we have a system of mutually interpenetrating hadrons (collision volume of the 2 heavy nuclei), each quark will find in its vicinity, at a distance less than the hadron radius, a number of quarks. This makes a macroscopic system of quarks and gluons.

(2) A macroscopic system is generally characterized by some state variables: number density ( $n$ ), pressure ( $P$ ), energy density ( $\epsilon$ ), temperature ( $T$ ) etc. This is not that simple! But let us assume this for the moment...

(3) At low density, a particular quark in a hadron 'knows' its partner quarks. However, at high density, when the hadrons start to interpenetrate each other, a particular quark will not be able to identify the quark which was its partner at lower density. Similar phenomena can happen at high temperature. As  $T$  is increased, more and more low mass hadrons (mostly pions) will be created. The system again will be dense enough and hadrons will start to interpenetrate. The system where, hadrons interpenetrate is best considered as a Quark matter, rather than made of hadrons.

(4) It is customary to call the quark matter as Quark-Gluon-Plasma (QGP). We define QGP as a thermalized state of quarks and gluons, where quarks and gluons are free to move over a nuclear volume rather than a nucleonic volume: it corresponds to  $T \simeq 200 \text{ MeV}$  or  $\epsilon \simeq 1 \text{ GeV/fm}^3$ .

---

QGP is the de-confined state of strongly interacting mater.

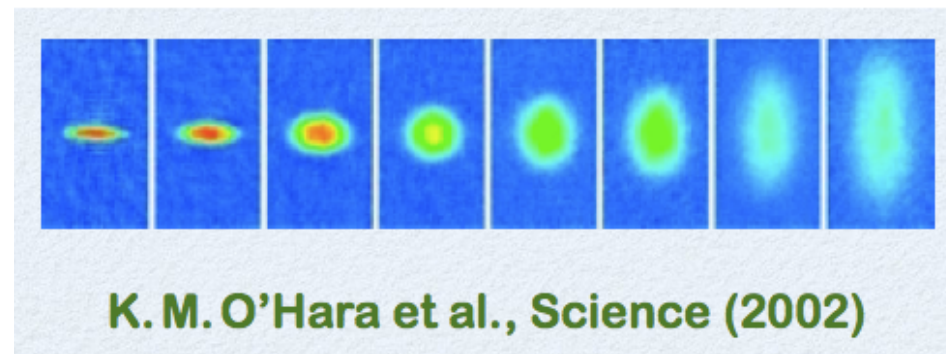
Since at low density or low  $T$  quarks are confined within the hadrons and at high  $n$  or at high  $T$ , quarks are de-confined, we can talk about a confinement/de-confinement transition. *This is not a phase transition.*

Quantitatively: Let us express the QCD potential as:  $V(r) = -\frac{\alpha}{r} + \sigma r$  for distances up to a few fm. And repulsive at very small distance  $< 1/n^3$ . Most of the transition above can be understood easily...

## Evolution: what we know from fermi liquids

---

This (ellipsoid) shape where a lot of particles are mixed is not a unique feature to heavy nuclei collisions. It happens also in the context of ultra-cold fermi liquids. Let us see what happens then.



Conclusion: The momentum expansion is along the small axis (of the initial shape in space).

## Position of the problem

---

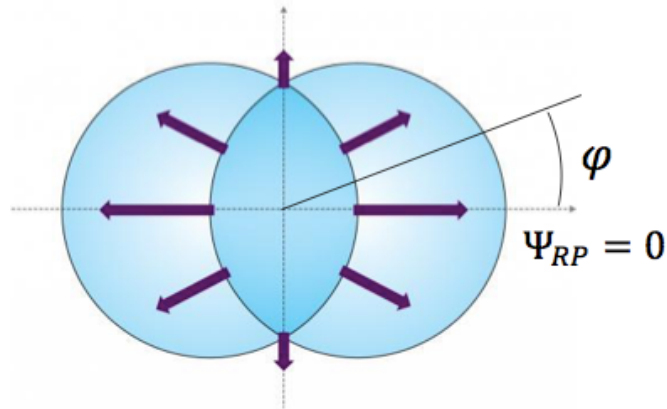
(1) A large number of constituents are mixed (QGP or not): the idea is then to use statistical physics. This is very unclear what this means in practice: can we use Maxwell-Boltzmann distributions, Fermi distributions, the notions of pressure, temperature etc. ?

Here, we make another assumption: that the mean free path of the constituents in the collision volume is much smaller than the typical size of the volume. This implies that there is enough interactions inside the interaction volume for a (local) thermal equilibrium to be reached.

Then,  $n$ ,  $P$  and  $T$  can be defined.

(2) The interaction volume (ellipsoid) presents an eccentricity: there is a small axis (along the reaction plane) and a large axis (perpendicular to the reaction plane).

Obviously, the gradients of any function along the small axis will be larger than the gradients along the large axis. Let us see (next) what this implies.



We write more formally the equation of motion for a small element of volume (of mass  $\rho.dV$ ) in the interaction volume:  
In a non-relativistic limit (Euler-Navier-Stokes equation), we get:

$$\frac{\partial}{\partial t} \vec{v} + (\vec{v} \cdot \nabla) \vec{v} = -\frac{1}{\rho} \nabla P + \frac{\eta}{\rho} \Delta \vec{v}.$$

We will neglect the viscosity and all quadratic terms for the velocities. In the relativistic limit, we obtain:

$$\frac{\partial}{\partial t} \vec{v} \simeq -\frac{1}{\epsilon + P} \left[ \nabla P + \vec{v} \frac{\partial P}{\partial t} \right].$$

The differences (relativistic limit or not) are not essential at this point of the discussion...

---

$$\frac{\partial}{\partial t} \vec{v} + (\vec{v} \cdot \nabla) \vec{v} = -\frac{1}{\rho} \nabla P.$$

This means that the pressure gradients drive the momentum evolution.

Therefore, the shape expands (in momentum) along the direction of the small initial axis.

This explains also the evolution of the shape in the Fermi liquid context.

## Is it correct?

---

This is correct when the pressure gradients drive the momentum evolution. This means under the assumption that the fluid mechanics equation(s) apply.

Stated differently: this is correct under the assumption that  $\lambda_{mfp} \ll L$ . Assumption that we have made above. As we have no way to check this, we can simply see if the observations are not too far from what we can extract from fluid mechanics.

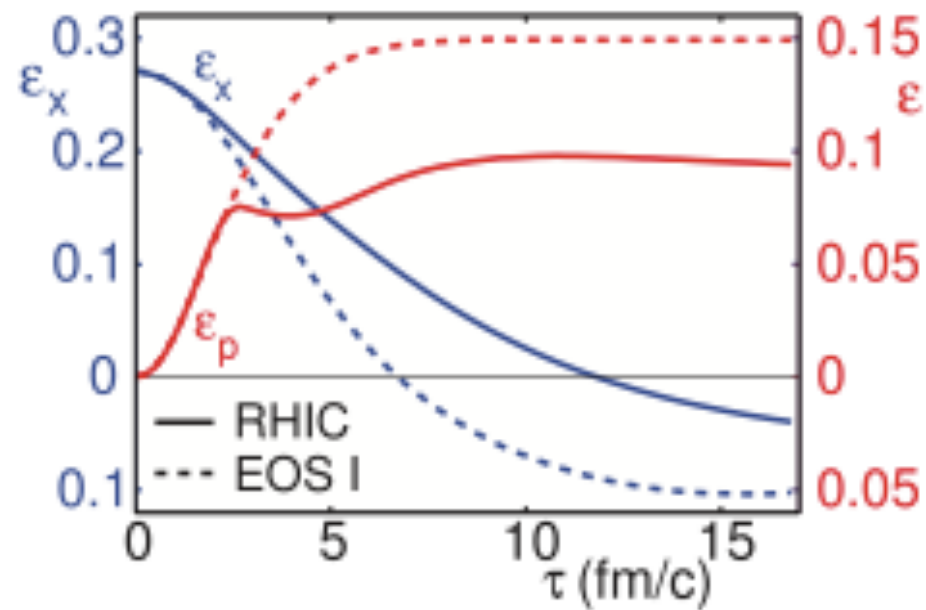
And this is the idea of everything related to heavy nuclei collisions.

There are plenty of reasons why this could not work:

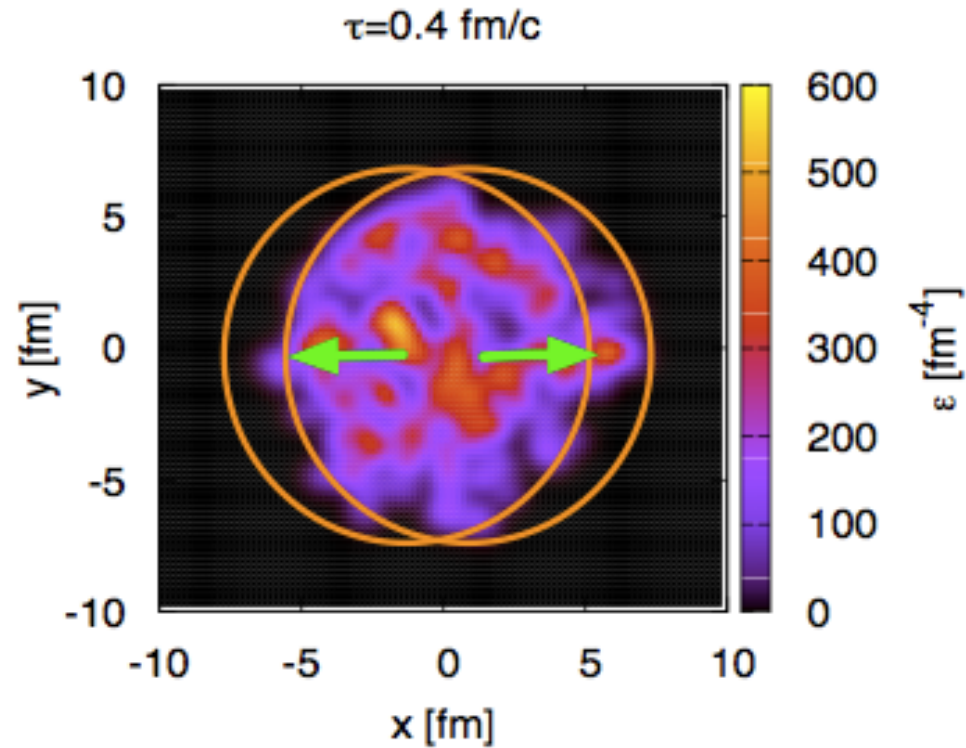
For example, if fundamental principles would imply that the medium can effectively reach thermal equilibrium but in a  $\delta t$  too long w.r.t. the collision time scales, this would not work.

Even more fundamentally, absolutely nothing guarantees that the laws of fluid mechanics apply at so small scale.

## A very first check



From P. F. Kolb and U. W. Heinz, In *\*Hwa et al.: Quark gluon plasma\** 634-714 [nucl-th/0305084]



(1) We assume that we have something like above which is formed (in the interaction volume) and (2) it expands in momentum like a fluid ( $\lambda_{mfp} \ll L$ , with  $\vec{v}(\vec{r}, t)$ ,  $P(\vec{r}, t)$  and  $T(\vec{r}, t)$  defined)...

## Before going further: what about $P, T, n...$ ?

---

If we want to reach a minimum of rationality, we need to define all the thermodynamical quantities correctly inside the collision volume.

This looks complicated but this is not. First, as the particle number is not fixed, the chemical potential is zero. Next, we can compute any distributions (Maxwell, Fermi or Bose), this will not change the fundamentals... Let us do it for massless bosons (the most simple):

$$n = \int \frac{d^3p}{h^3} \frac{1}{e^{pc/(k_B T)} - 1} = \frac{8\pi\zeta(3)}{h^3} T^3, \quad \zeta(3) \simeq 1.2.$$

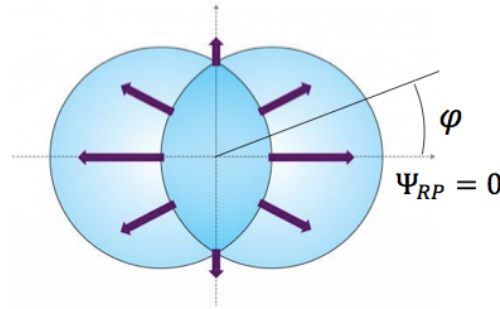
$$\epsilon = \int \frac{d^3p}{h^3} \frac{pc}{e^{pc/(k_B T)} - 1} = \frac{4\pi^5}{15h^3} T^4.$$

$$P = \int \frac{d^3p}{h^3} \frac{pc/3}{e^{pc/(k_B T)} - 1} = \epsilon/3.$$

And finally, the entropy:  $s = (\epsilon + P)/T \simeq 3.6n$ .

For the other distributions, all these results would be modified by a scaling factor close to 1. For example, for fermions, the entropy would be found as:  $s = (\epsilon + P)/T \simeq 4n$ .

## From fluid mechanics to collectivity



$\phi$  labels the azimuthal angle of the momentum of a given particle (expanding) within the medium (interaction volume) w.r.t. the interaction plane (the reaction plane ( $\vec{b}, \vec{e}_z$ ) is assumed to exist and follow the equation  $\psi_{RP} = 0$ ).

Then, it is possible to write a Fourier series of the  $\phi$  distribution of the expanding particles as:

$$\frac{dN}{d(\phi - \psi_{RP})} = \frac{N}{2\pi} \left[ 1 + 2 \sum v_n \cos(n(\phi - \psi_{RP})) \right]$$

The sum is over  $n$  (the harmonic number).

We can simplify:

$$\frac{dN}{d\phi} = \frac{N}{2\pi} \left[ 1 + 2 \sum v_n \cos(n(\phi - \psi_{RP})) \right]$$

## Some comments

---

$$\frac{dN}{d\phi} = \frac{N}{2\pi} \left[ 1 + 2 \sum v_n \cos(n(\phi - \psi_{RP})) \right]$$

Then:

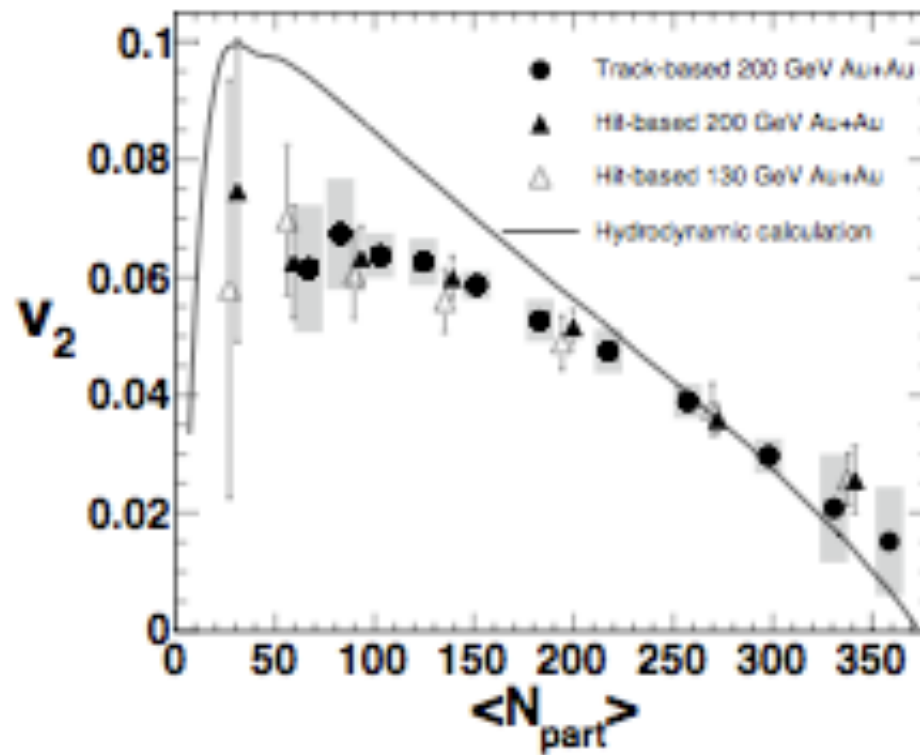
$$v_n = \langle \cos(n(\phi - \psi_{RP})) \rangle = \frac{\int d\phi \frac{dN}{d\phi} \cos(n(\phi - \psi_{RP}))}{\int d\phi \frac{dN}{d\phi}}.$$

Where the average is taken over all particles and for all events at a given centrality. This means that we assume that  $\psi_{RP}$  is -almost- the same for all the events considered.

For example,  $v_2 = \langle \cos(2(\phi - \psi_{RP})) \rangle$  is the first important potentially non-zero quantity to check if all this makes sense.

What we expect is that  $v_2 > 0$  for non-central collisions, where the ellipsoid shape can exist in the initial stage and is found  $v_2 \simeq 0$  for central collisions where there is no spatial anisotropy in the initial stage...

*not too bad!*



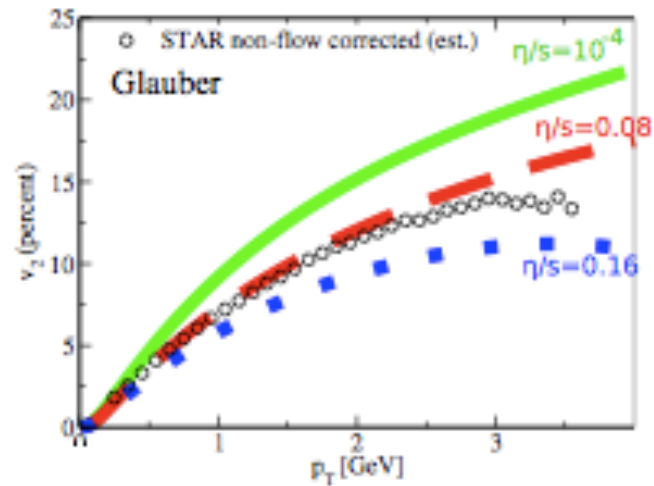
PHOBOS in 2004

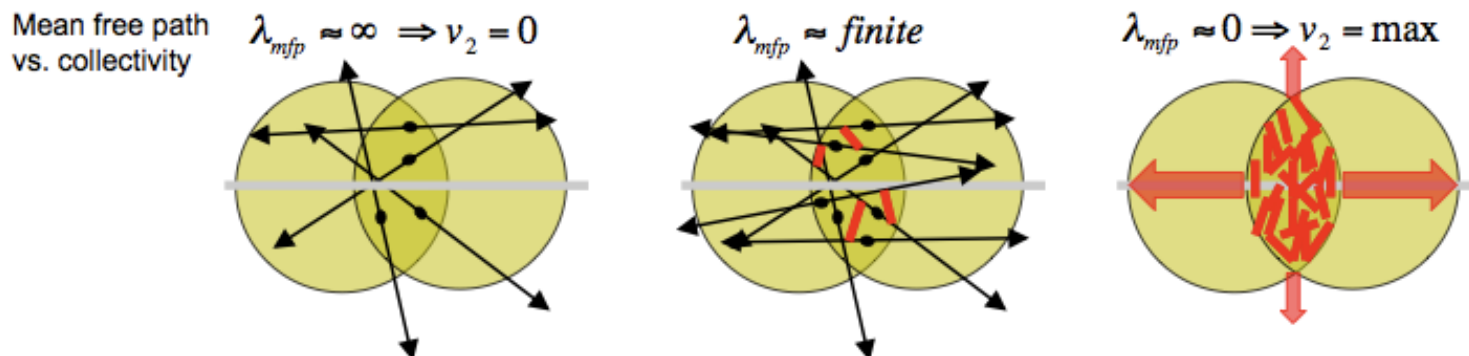
---

In general, coefficients  $v_n$  may be functions of  $p_T$  and  $y$ :

$$E \frac{d^3 N}{d^3 p} = \frac{d^2 N}{2\pi p_T dp_T dy} \left[ 1 + 2 \sum v_n(.) \cos(n(\phi - \psi_{RP})) \right]$$

Then, this is possible to test some models, for example on the observable  $v_2(p_T)$ :





It is time to come back on some assumptions: on the existence and properties of the reaction plane (RP).

(0) Obviously, only in the context of picture (3), there can be a RP and thus non-zero  $v_n$  coefficients.

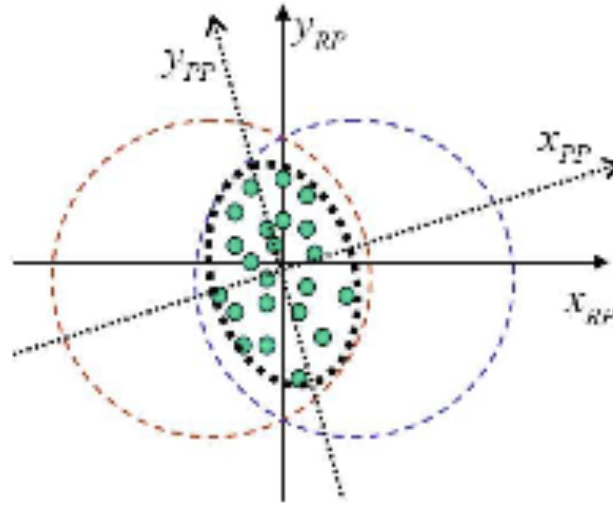
(1) We have not shown whether this is possible or not to determine (experimentally) the reaction plane!

*Indeed, this is possible to determine  $v_2$  without the experimental knowledge of the RP.*

(2) Moreover,  $\psi_{RP}$  depends on  $n$  (the harmonic index) and will change (fluctuate) event by event.

## Fluctuation of $\psi_{RP}$ event by event

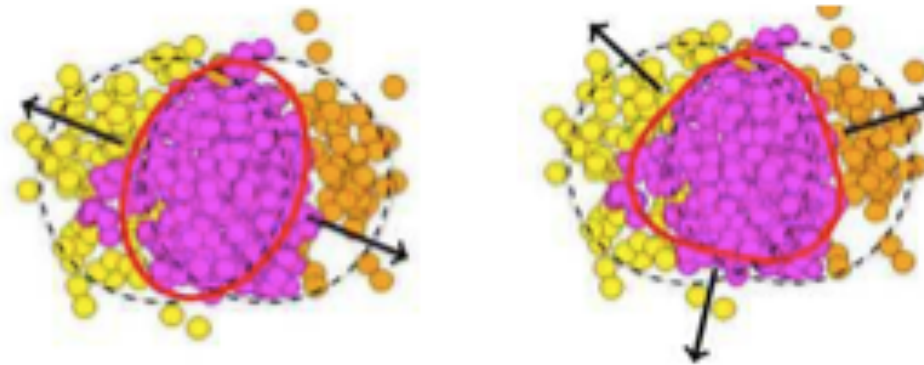
---



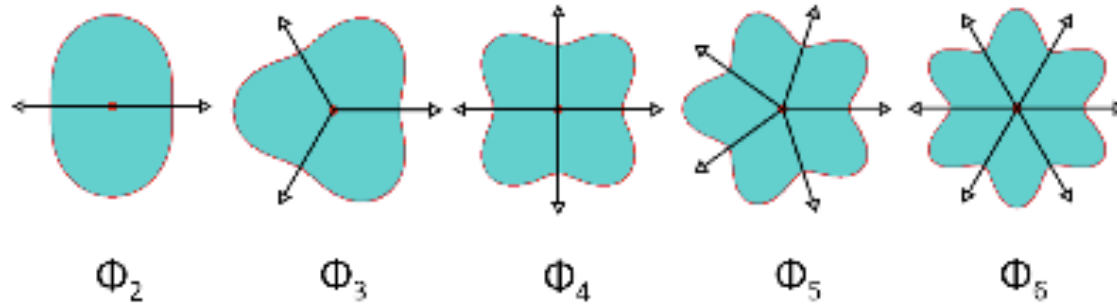
Clearly, the physical reference angle is  $\Phi_{PP}$  (PP for participant-plane) (which is not  $\psi_{RP}$  for a particular event). We will see later what this implies on the harmonic coefficients  $v_n$ .

$$\psi_{RP}[n]$$


---



What enters in the harmonic series are terms in  $\cos(n(\phi - \psi_{RP}))$ , where  $\phi - \psi_{RP}$  refers to the angle w.r.t. a reference axis for the expansion. But, this depends on the initial shape... Till now, we have simplified the discussion using only ellipsoid shape: **depending on  $n$ , we see that the expansion depends on  $n$  (figure).**



$$\frac{dN}{d\phi} = \frac{N}{2\pi} \left[ 1 + 2 \sum v_n \cos(n(\phi - \Phi_n)) \right]$$

$$v_n = \langle \cos(n(\phi - \Phi_n)) \rangle = \frac{\int d\phi \frac{dN}{d\phi} \cos(n(\phi - \Phi_n))}{\int d\phi \frac{dN}{d\phi}}.$$

Angles  $\Phi_n$  are the directions of maximum emission in momentum (always along the small axis)

We keep the notation  $\Phi$  for the PP (participant plane), which differs on an event/event basis from the theoretical RP.

## In practice

---

Let us compute:  $\langle \cos(n(\phi_1 - \phi_2)) \rangle$  (2-particles correlation) averaged over all particles (pairs) in one event and over all events.

$$\langle \cos(n(\phi_1 - \phi_2)) \rangle = \langle \cos(n(\phi_1 - \Phi_n)) \rangle \langle \cos(n(\phi_2 - \Phi_n)) \rangle + \dots$$

where the ... stands for the non-flow term (we neglect it here).

$$\langle \cos(n(\phi_1 - \phi_2)) \rangle = \langle v_n^2 \rangle$$

where  $\langle v_n^2 \rangle$  is the average over all events.

Remember that we have seen that the RP fluctuates event by event. This means that  $\langle v_n^2 \rangle \neq \langle v_n \rangle^2$ .

The difference between these 2 terms is the part related to the fluctuation of the the RP event/event:

$$\langle v_n^2 \rangle = \langle v_n \rangle^2 + \sigma_n^2.$$

---

What is measured is:  $v_n(\{2\}) = \sqrt{\langle \cos(n(\phi_1 - \phi_2)) \rangle}$ . Then, following the logic of the previous slide, we can derive from this  $\langle v_n \rangle$  up to a fluctuation term.

$$v_n(\{2\}) = \langle v_n \rangle + \frac{\sigma_n^2}{2 \langle v_n \rangle}$$

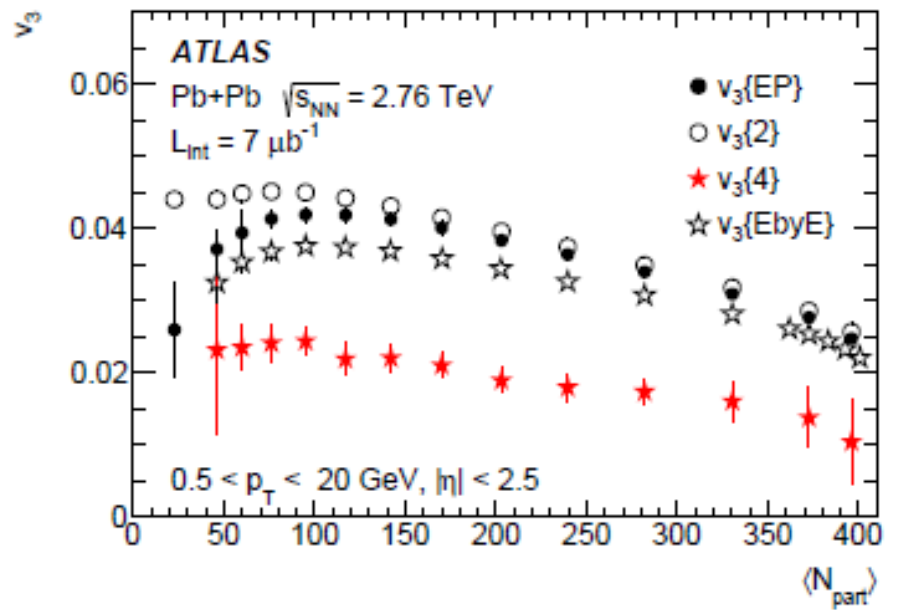
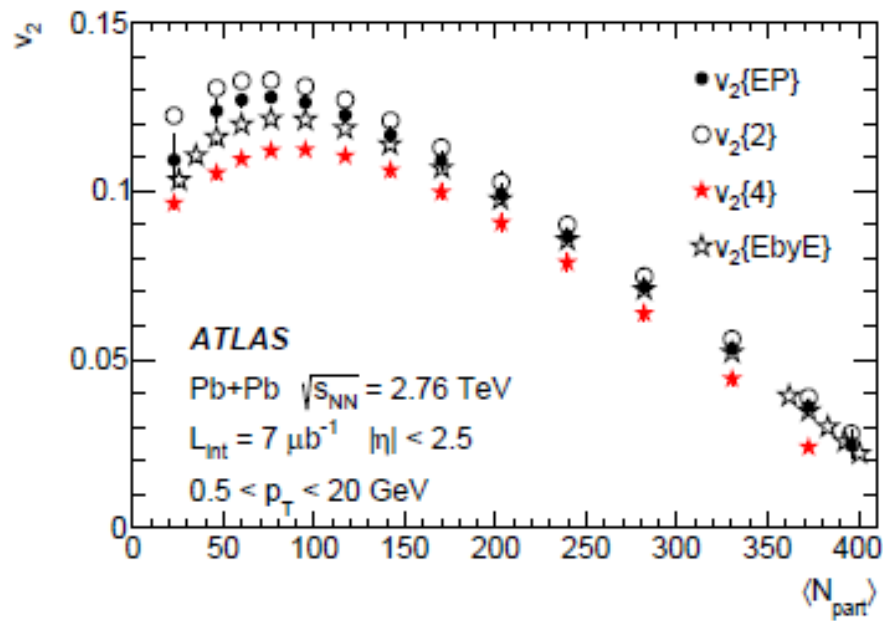
But, this is not the end of the story. We can also measure 4-particles correlations:

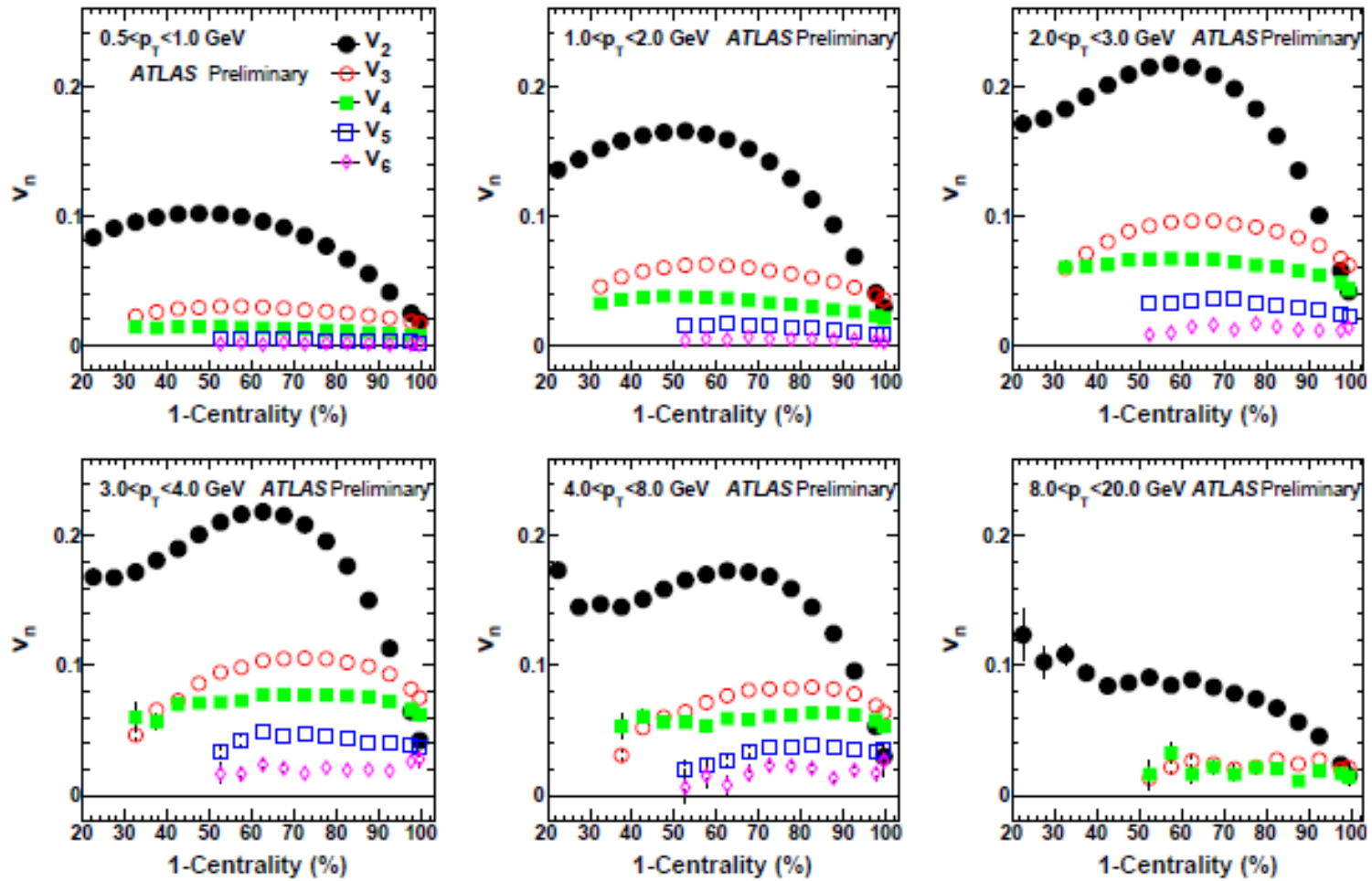
$$v_n(\{4\}) = \sqrt{\langle \cos(n(\phi_1 + \phi_2 - \phi_3 - \phi_4)) \rangle}.$$

With the same method, we can show:

$$v_n(\{4\}) = \langle v_n \rangle - \frac{\sigma_n^2}{2 \langle v_n \rangle}.$$

Interestingly, we can eliminate the fluctuation term combining both relations.





## Let us come back to the ideas on expansion (of the ellipsoid)

---

Our key point is that the ellipsoid shape (in space) leads to large gradients along the small axis, which are transferred to momentum distributions BECAUSE the mixture inside the interaction volume is behaving like a fluid (following fluid mechanics).

Therefore, if we quantify the asymmetry in space (of the collision volume) as:

$$\epsilon_2 = \frac{\langle y^2 \rangle - \langle x^2 \rangle}{\langle y^2 \rangle + \langle x^2 \rangle},$$

then, qualitatively  $\epsilon_2 \propto v_2 \dots$

It would be interesting to push this idea beyond its qualitative expression.

Also, if we compute  $\Phi_2^*$  the reference angle for the ellipsoid (using the space coordinates of particles in the interaction volume), then, obviously  $\Phi_2^* \equiv \Phi_2$  (following the above discussion). We must notice that these 2 angles are different: one includes particle coordinates and the other is related to momentum expansion.

## Position of the problem

---

In fact, this will be more complicated for higher harmonics. In general, we do not have:  $\epsilon_n \propto v_n$  and  $\Phi_n^* \equiv \Phi_n$  for harmonics larger than 2.

However, that is also clear is that the flow expansion leads to correlations between the space observables and the momentum ones.

First, let us write the generic formula for  $\epsilon_n$ :

$$\epsilon_n = \frac{\sqrt{\langle r^n \cos(n\phi) \rangle^2 + \langle r^n \sin(n\phi) \rangle^2}}{\langle r^n \rangle},$$

where  $r, \phi$  are the polar transverse coordinates of the particles in the interaction volume and the average is taken over all these particles. Similarly:

$$n\Phi_n^* = \arctan \frac{\langle r^n \sin(n\phi) \rangle}{\langle r^n \cos(n\phi) \rangle} \bmod(\pi)$$

---

What can be the extension of the first-idea ?  $v_n e^{in\Phi_n} \propto \epsilon_n e^{in\Phi_n^*}$

For  $n = 2, 3$  this is correct. For larger harmonics, for example  $n = 4$ , we can imagine that  $v_4$  receives contribution from  $\epsilon_4$  but also from terms in  $\epsilon_2^2$ . We can write this possibility as:

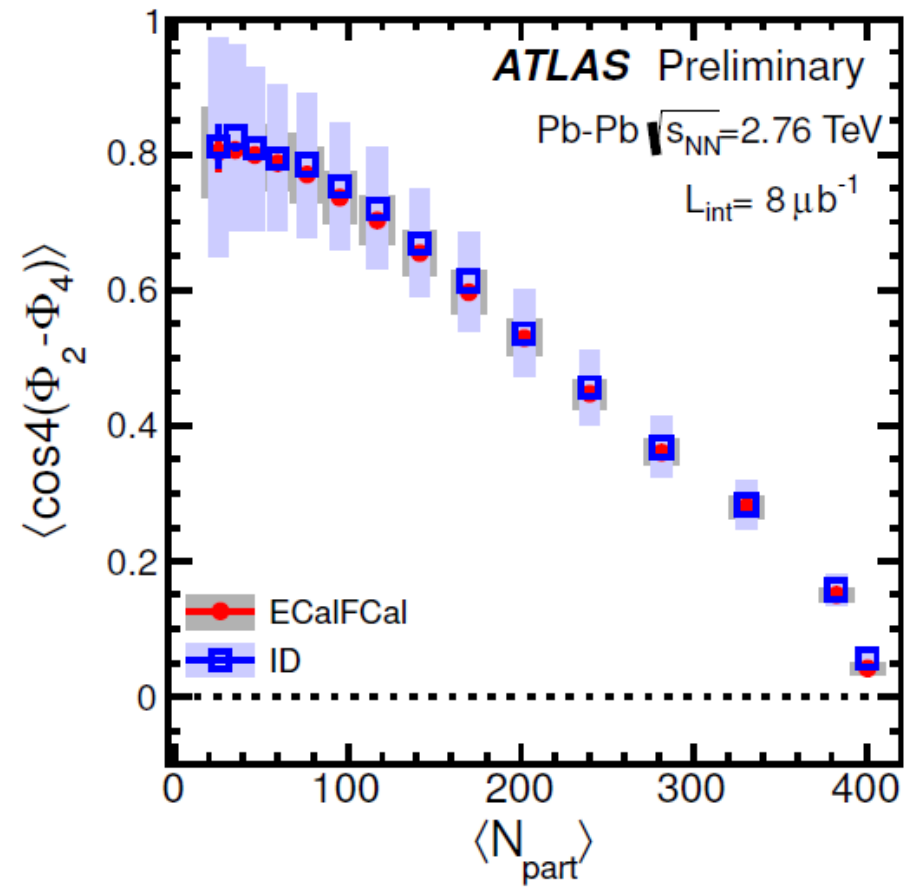
$$v_4 e^{i4\Phi_4} = \alpha \epsilon_4 e^{i4\Phi_4^*} + \alpha' \epsilon_2^2 e^{i4\Phi_2^*}.$$

Similarly, for  $v_5 e^{i5\Phi_5}$ , we could write:

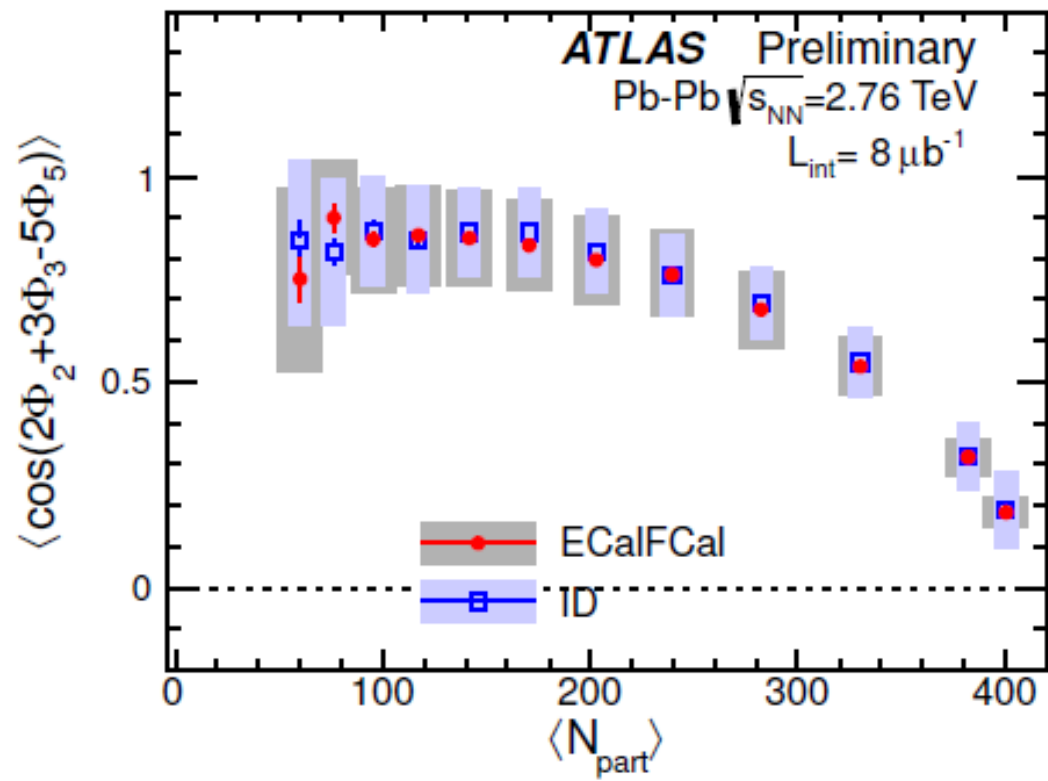
$$v_5 e^{i5\Phi_5} = \beta \epsilon_5 e^{i5\Phi_5^*} + \beta' \epsilon_2 e^{i2\Phi_2^*} \epsilon_3 e^{i3\Phi_3^*}.$$

Of course, as  $\epsilon_{2,3} e^{i(2,3)\Phi_{2,3}^*}$  are proportionnal to  $v_{2,3} e^{i(2,3)\Phi_{2,3}}$ , we can make the replacement in the right hand side of the above equations.

If this is correct, this means that:  $\epsilon_2$  drives both  $v_2$  and  $v_4$  and thus correlations between  $\Phi_2$  and  $\Phi_4$  should be expected. More precisely:  $\langle \cos(4\Phi_2 - 4\Phi_4) \rangle$  averaged over all events at a given centrality should not be zero.



It looks like it works... *Note: it has never been done by CMS.*



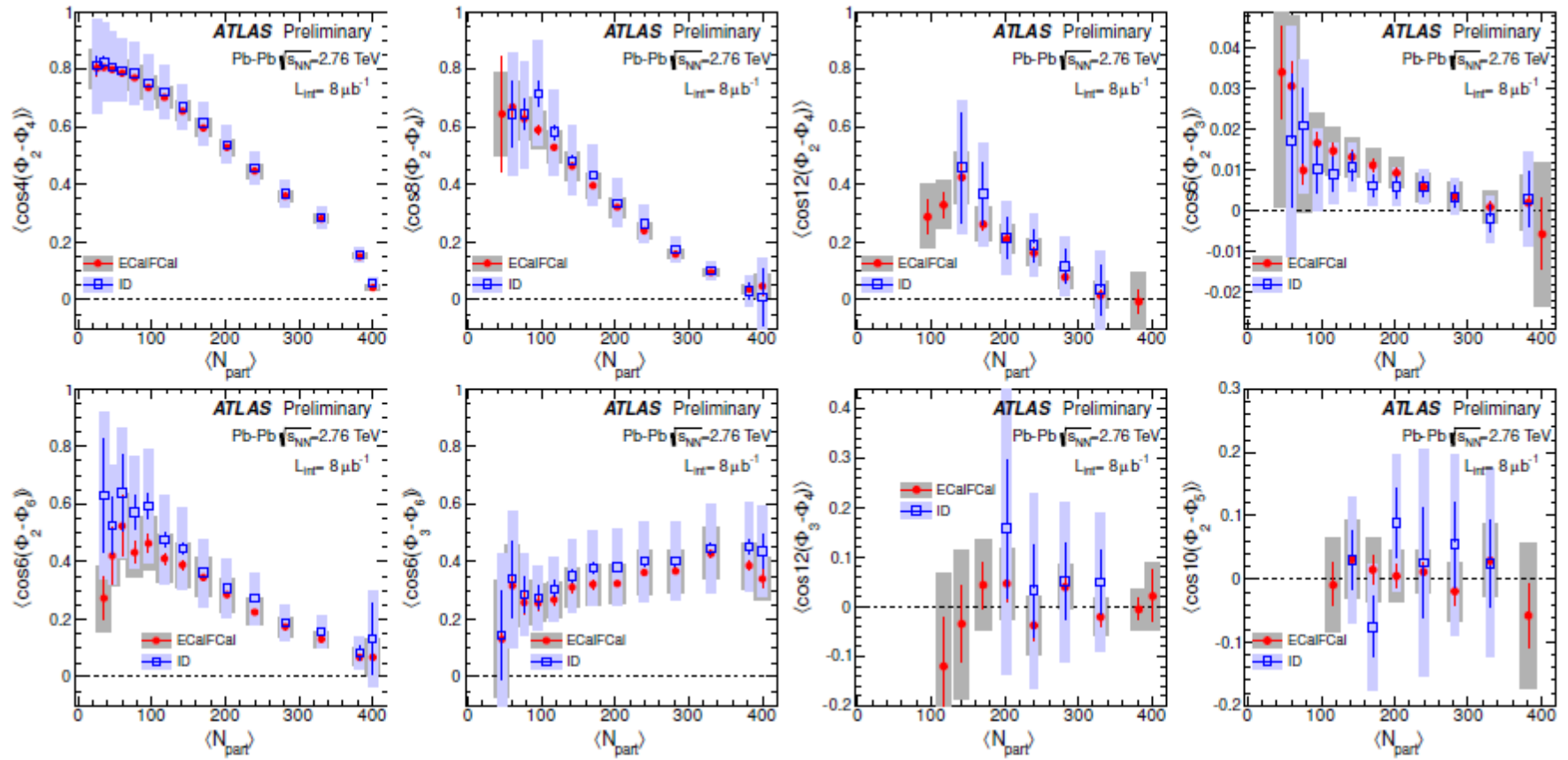
---

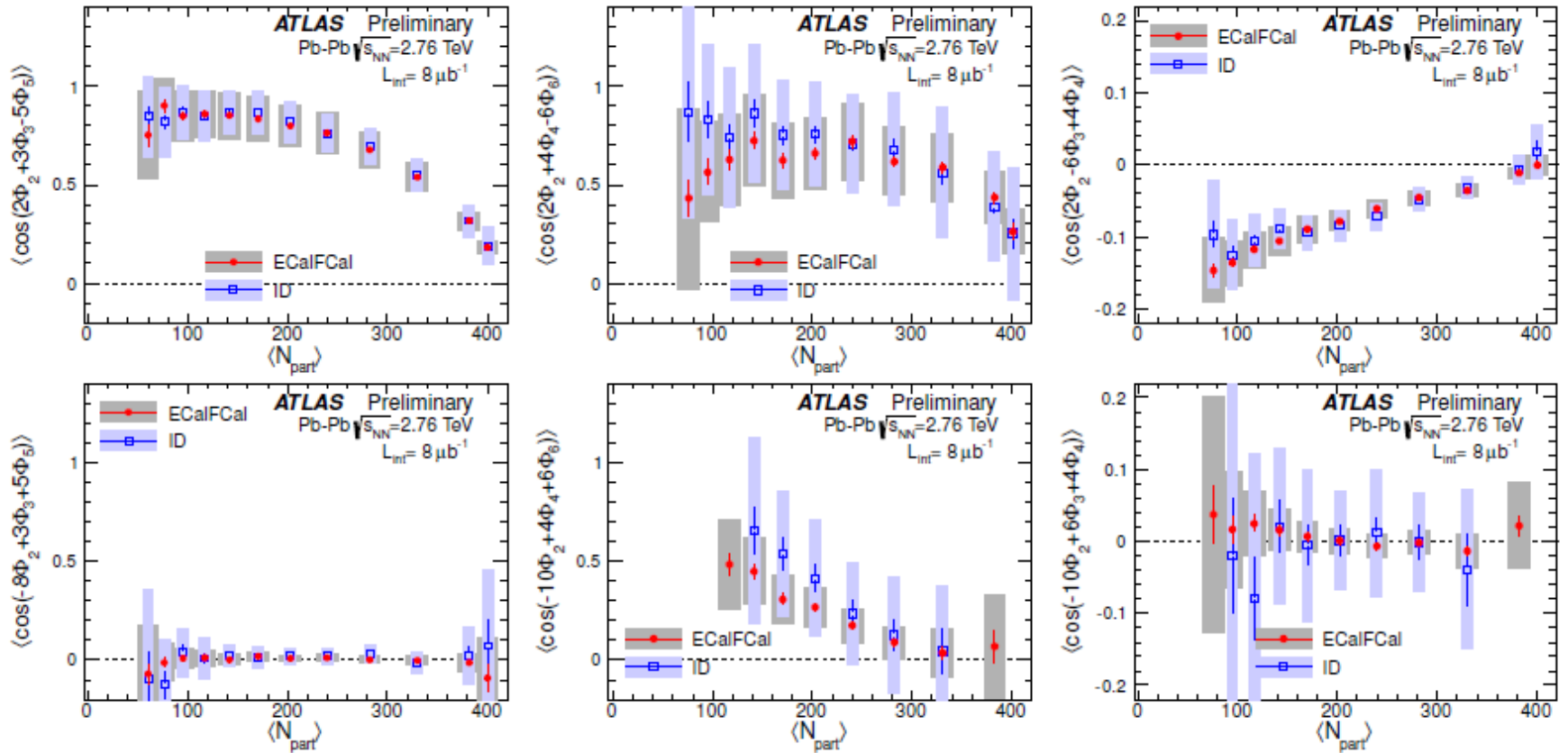
$\langle \cos(4\Phi_2 - 4\Phi_4) \rangle$  and  $\langle \cos(2\Phi_2 + 3\Phi_3 - 5\Phi_5) \rangle$  are non-zero.

Confirming that the relations:  $v_4 e^{i4\Phi_4} = \alpha \epsilon_4 e^{i4\Phi_4^*} + \alpha'' v_2^2 e^{i4\Phi_2}$  and  $v_5 e^{i5\Phi_5} = \beta \epsilon_5 e^{i5\Phi_5^*} + \beta'' v_2 e^{i2\Phi_2} v_3 e^{i3\Phi_3}$  are reasonable.

(1) Here we have assumed that the higher harmonics in the flow coefficients can receive contributions from different terms in the initial geometry... Then, the above equations are an heuristic way to write a phenomenology of the hydrodynamic response of the collective flow. The corrective terms are non-linear.

(2) However, even if we stick to a purely linear response, we could imagine that there are correlations between the different  $\Phi_n^*$  in the initial geometry of the collision volume, which are subsequently transferred to correlations in the  $\Phi_n$ .





---

## Chapter II

With the equations of fluid mechanics

## Fluid mechanics: reminder

---

Let us start with the Navier-Stokes equation(s):

$$\frac{\partial}{\partial t} \vec{v} + (\vec{v} \cdot \nabla) \vec{v} = -\frac{1}{\rho_0} \nabla P + \frac{\eta}{\rho_0} \Delta \vec{v}.$$

This is only correct for incompressible fluid... that we will always assume: this is synonymous of expansion in the vacuum.

Only at this condition the shear viscosity has the form above. We neglect the viscosity as a first approximation and write the equation only for the  $x$  and  $y$  components in the transverse plane:

$$\frac{\partial v_x}{\partial t} + \left[ v_x \frac{\partial}{\partial x} + v_y \frac{\partial}{\partial y} \right] v_x = -\frac{1}{\rho_0} \frac{\partial P}{\partial x};$$

$$\frac{\partial v_y}{\partial t} + \left[ v_x \frac{\partial}{\partial x} + v_y \frac{\partial}{\partial y} \right] v_y = -\frac{1}{\rho_0} \frac{\partial P}{\partial y}.$$

---

At this level, we can enter the scales (small axis  $x$  and large axis  $y$ ) and make the corresponding approximations.

One important question: do we need a relativistic formulation for the motion in the transverse plane?

The answer that we find in literature, articles etc. is: Yes. Thus, we will do it... However, this is not that clear *to my view*.

Proof: let us write the energy of the system of particles that are moving very fast along the  $z$  axis (we approximate the very large momentum  $p_z$  to be the same for all constituents):

$$E = \sum_i \sqrt{p_z^2 + p_{x,i}^2 + p_{y,i}^2 + m_i^2}.$$

But  $p_z$  is the large component, thus we can develop the expression above as:

$$E = \sum_i p_z + \frac{p_{x,i}^2 + p_{y,i}^2}{2p_z} + \frac{m_i^2}{2p_z},$$

where  $\sum_i p_z$  is the  $P_z$  of the system (conserved quantity).

---

$$E - P_z = \sum_i \frac{p_{x,i}^2 + p_{y,i}^2}{2p_z} + \frac{m_i^2}{2p_z}$$

is a non-relativistic expression for the energy, with the inertia plays by the  $p_z$ .

This means that when we consider a system with a very large boost against the  $z$  axis (w.r.t. the motion in other directions), then, in the transverse plane (which means  $x, y$  plane) the motion of the system is non-relativistic.

However, we need to expose the relativistic version of fluid mechanics (because this is what is used in practice for the expansion in the transverse plane)... and, anyway, this is useful to describe the longitudinal behaviour when needed.

We will restrict the presentation to the ideal fluids (with no viscosity) and just consider viscosity at the end as a correction to the ideal case.

# Relativistic fluid mechanics

---

It is useful to define the 4-velocity  $u^\mu$ :  $u^0 = \gamma$  and  $\vec{u} = \gamma\vec{v}$ . Then, as usual in relativity, equations of motion are written as:

$$\partial_\mu T^{\mu\nu} = 0,$$

where  $T^{\mu\nu}$  is the energy-momentum tensor. For ideal fluids, it is given by

$$T^{\mu\nu} = (\epsilon + P)u^\mu u^\nu - P g^{\mu\nu},$$

where  $g^{\mu\nu} \equiv \text{diag}(1, -1, -1, -1)$  and where  $\epsilon$  is the internal energy density per unit volume, which includes the mass energy density in the relativistic formulation.

The different components of the energy-momentum tensor can be understood as follows:

- $T^{00}$  is the energy density;
- $T^{0j}$  is the density of the  $j^{th}$  component of momentum, with  $j = 1, 2, 3$ ;
- $T^{i0}$  is the energy flux along axis  $i$ ;
- $T^{ij}$  is the flux along axis  $i$  of the  $j^{th}$  component of momentum.

---

We can easily check that  $T^{\mu\nu}$  reduces to a simple matrix  $T$  in the rest frame of the fluid, where  $u^\mu = (1, 0, 0, 0)$ . We obtain

$$T = \begin{pmatrix} \epsilon & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{pmatrix}$$

Indeed, in the fluid rest frame, the assumption of local thermodynamic equilibrium implies isotropy. Hence, the energy flux  $T^{i0}$  and the momentum density  $T^{0j}$  vanish. In addition, this implies that the pressure tensor is proportional to the identity matrix,  $T^{ij} = P\delta_{ij}$ .

Then, the general form for  $T^{\mu\nu}$  is obtained with Lorentz transformations.

Let us rewrite  $\partial_\mu T^{\mu\nu} = 0$ :

$$\begin{aligned} \frac{\partial P}{\partial t} - \frac{\partial}{\partial t} \left( \frac{\epsilon + P}{1 - \vec{v}^2} \right) - \nabla \cdot \left( \frac{(\epsilon + P)\vec{v}}{1 - \vec{v}^2} \right) &= 0, \\ \frac{\partial}{\partial t} \vec{v} + (\vec{v} \cdot \nabla) \vec{v} &= -\frac{1 - \vec{v}^2}{\epsilon + P} \left[ \nabla P + \vec{v} \frac{\partial P}{\partial t} \right]. \end{aligned}$$

The first equation is for energy conservation, the second one for the equation of motion ( $\sim$  Euler equation). Of course, in addition, we need to consider the law of thermodynamics that make the link between  $P$ ,  $T$  and  $\epsilon$ .

---

Using the notation  $D = u^\mu \partial_\mu$ , these equations can be written in a more compact form as

$$D\epsilon = -(\epsilon + P)\nabla\vec{u},$$

$$Du^\nu = -\frac{\nabla^\nu P}{(\epsilon + P)}.$$

# The transverse motion in relativistic fluid mechanics

---

Let us come back on the equations of motion in the transverse plane (where we neglect the quadratic terms):

$$\frac{\partial v_x}{\partial t} = -\frac{1}{\epsilon + P} \frac{\partial P}{\partial x};$$

$$\frac{\partial v_y}{\partial t} = -\frac{1}{\epsilon + P} \frac{\partial P}{\partial y};$$

With:

$$\frac{1}{\epsilon + P} \frac{\partial P}{\partial x} = c_s^2 \frac{\partial \log s}{\partial x}.$$

Similarly for the  $y$  component.

Where  $c_s$  is the velocity of sound in the fluid:  $c_s^2 = \frac{\partial P}{\partial \epsilon}$  and  $s$  is the entropy density:  $\epsilon = -P + Ts$  (and  $dP = sdT$ ,  $d\epsilon = Tds$ ).

We need to have an idea of  $s(x, y)$ . We will assume that it has a Gaussian form with r.m.s. of a few fm.

Also, we can safely assume that the initial transverse velocities are (almost) zero. This is needed to integrate the equations.

Then:  $v_x = c_s^2 x t / \sigma_x^2$  and  $v_y = c_s^2 y t / \sigma_y^2$ .

## Comments (transverse expansion)

---

We have shown:  $v_x = c_s^2 x t / \sigma_x^2$  and  $v_y = c_s^2 y t / \sigma_y^2$ .

The transverse expansion is a smooth process (linear in space and time), despite the very large gradients. This is due to the fact that the inertia  $P + \epsilon$  is also large.

If  $x$  is the short axis, then:  $\sigma_x < \sigma_y$  and this implies that the expansion is faster along the short axis (as already stated).

And, this results in more particles emitted near  $\phi = 0$  and  $\phi = \pi$ , i.e., parallel to the  $x$ -axis, than near  $\phi = \pm\pi/2$ , parallel to the  $y$ -axis. This effect corresponds to the  $\cos 2\phi$  term in the Fourier decomposition of the azimuthal distribution.

# The longitudinal motion in relativistic fluid mechanics

---

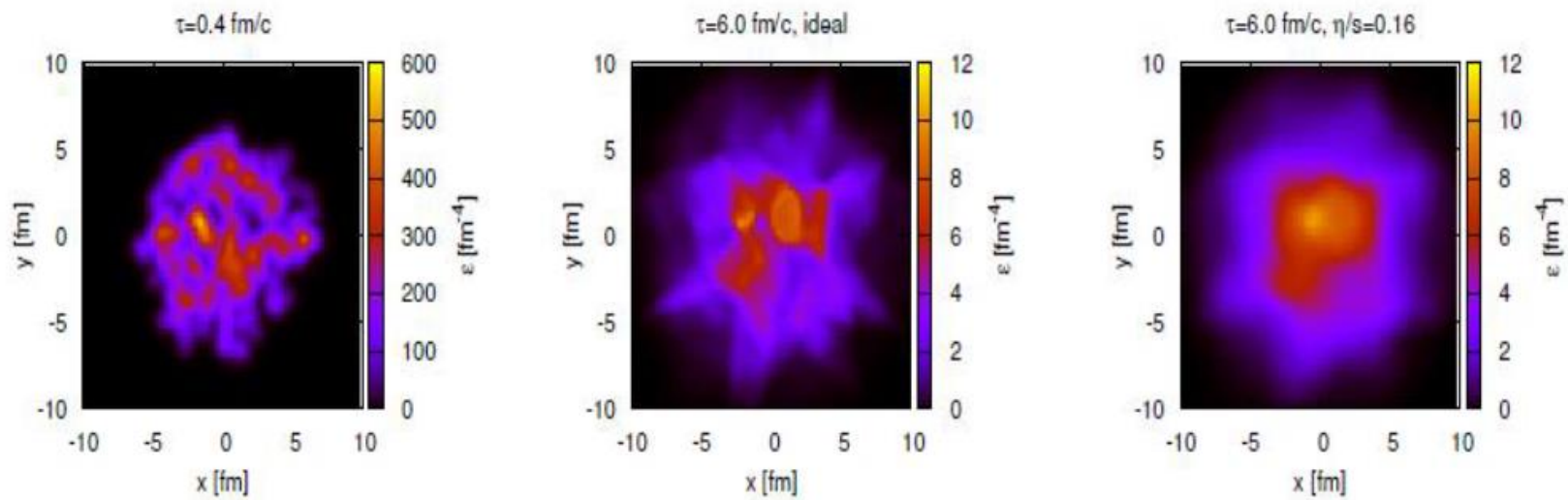
Same method as before:

$$\frac{\partial v_z}{\partial t} = -\frac{1}{\epsilon + P} \frac{\partial P}{\partial z} = -c_s^2 \frac{\partial \log s}{\partial z}.$$

However, contrary to the motion in the transverse plane, the situation is more complex now. We do not know the initial conditions and it is not trivial to get an intuition on  $s(z)$ ...

This means that any continuation will be model dependent. There are several standard models to do it...

## The effect of viscosity (in picture)



## Viscosity: some open ideas

---

Viscosity is related to the ability of the fluid to return to local thermodynamic equilibrium. For a medium with strongly interacting constituents at the beginning of the hydrodynamical evolution, we can expect small relaxation times, thus small viscosity effects.

However, during the expansion, as the mean free path increases, the coupling strength is decreased between constituents. Therefore, viscosity effects could become large during the expansion.

Thus, viscosity effects should not be encoded once for all in the equations at the beginning (as an average effect). This is what is done in practice...

## A temporary conclusion

---

HI in ATLAS is a field at the border between classical mechanics, statistical physics, where one can think in terms of images in the real world.

This is interesting...

Moreover, this is clearly a field where new ideas would be useful:

Can we measure flow only with photons? and compare? Viscosity effects?

There are plenty of checks to do on non-flow effects (to be sure that present measurements are really what they are supposed to be).

Compare  $PbPb$ ,  $pPb$  and  $pp$  and try to understand what happens.

The most important: till now, nothing has been done on the essential quantity in fluid mechanics: the vorticity (next slides).

...

I have not discussed the hard probes... This is another story.

# Vorticity in fluid mechanics

---

In the non-relativistic limit, vorticity is defined from the velocity field:  $\vec{\omega} = \vec{\nabla} \times \vec{v}$ .

Therefore, a first guess would be: writing equations using  $\vec{v}$  (Euler, Navier-Stokes) or using the vorticity is equivalent. This is obviously correct...

However, the vorticity (as a physical quantity) is playing such an important role in fluid mechanics that it is very very useful to read the equations directly for it.

First, the basics: Euler equation for incompressible fluids leads to:  $\frac{\partial \vec{\omega}}{\partial t} = \vec{\nabla} \times (\vec{v} \times \vec{\omega})$ . This implies that  $\int \vec{v} \cdot d\vec{r}$  is conserved, where the integral is evaluated on a closed contour moving with the fluid (thus made of fluid particles) (Helmholtz-Kelvin theorem) ; and  $\int \vec{v} \cdot \vec{\omega} dV$  is also conserved, where the integral is computed over the all space.

Another idea: if we have a plate moving along the  $x$  axis in a fluid filling the full space, the plate is imposing a velocity field in  $v_x(y, t)$ . Then the vorticity is equal to:  $\omega_z = -\frac{\partial v_x}{\partial y}$  and the equation for the vorticity reads:

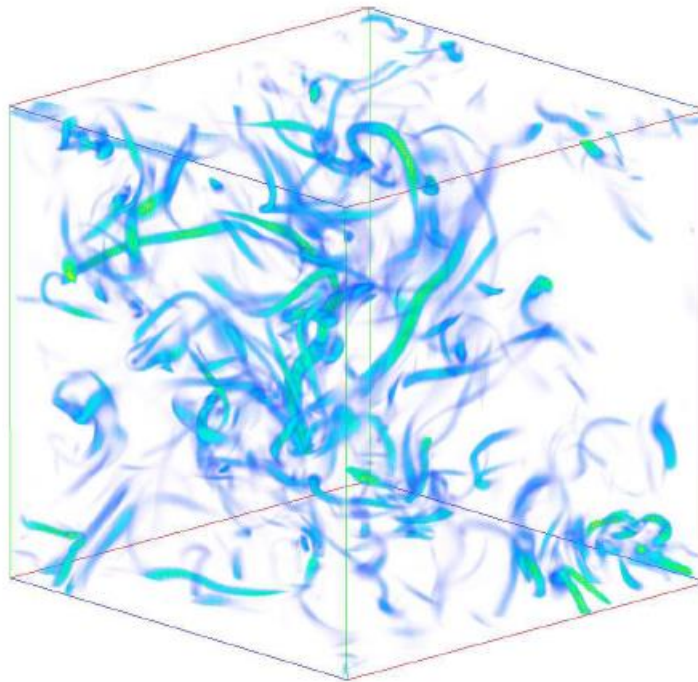
$$\frac{\partial \omega_z}{\partial t} = \frac{\eta}{\rho} \frac{\partial^2 \omega_z}{\partial y^2}.$$

This is simply a diffusion equation.

# The essence of fluid mechanics

---

How the vortex tubes formed in a fluid could behave on the long run?



## Vorticity and relativity

---

The immediate generalization of the standard definition reads:

$$\omega_{rel}^{\mu} = \epsilon^{\mu\nu\rho\sigma} u_{\nu} \partial_{\rho} u_{\sigma},$$

where  $u^{\mu}$  has been defined in chapter II. This gives:  $\vec{\omega}_{rel} = \gamma^2 \vec{\omega} + \gamma^2 \vec{v} \times \partial_t \vec{v}$  and  $\omega_{rel}^0 = \gamma^2 \vec{v} \cdot \vec{\omega}$ .

Therefore, if we neglect the terms of order 2 in the velocities, these definitions are very close to the non-relativistic ones. Thus, all ideas about vorticity that we can develop in standard cases can hold in the context of HI collisions.

In particular, the equation for the vorticity  $\vec{\omega}_{rel}$  (neglecting the viscosity) is the same as before:

$$\frac{\partial \vec{\omega}_{rel}}{\partial t} = \vec{\nabla} \times (\vec{v} \times \vec{\omega}_{rel}).$$

Let us examine what this gives for the expansion in the transverse plane...

---

Remember that:  $\frac{\partial \vec{\omega}_{rel}}{\partial t} = \vec{\nabla} \times (\vec{v} \times \vec{\omega}_{rel})$  leads to the theorem of Helmholtz-Kelvin.

This implies that the vortex lines are frozen in the fluid. Then, the vorticity will decay due to the expansion of the system.

Conclusion: we have had very easily an idea of what happens and how the vortex tubes will evolve in the system.

In equations. We have already shown that:  $v_x = c_s^2 x t / \sigma_x^2$  and  $v_y = c_s^2 y t / \sigma_y^2$ . We will also assume that  $v_z \simeq z/t$  is characteristic of the longitudinal expansion and that the vorticity is along the  $y$  axis. We can include all this in the equation of vorticity (above) and then solve in  $\omega_y$ . We obtain:

$$\omega_y(\vec{x}, t) = \frac{t_0}{t} \exp\left[-\frac{c_s^2}{2\sigma_x^2}(t^2 - t_0^2)\right] \omega_y(\vec{x}_0, t_0),$$

where  $\vec{x}$  and  $\vec{x}_0$  are related as:  $x = x_0 \exp\left[-\frac{c_s^2}{2\sigma_x^2}(t^2 - t_0^2)\right]$ ,  $y = y_0 \exp\left[-\frac{c_s^2}{2\sigma_y^2}(t^2 - t_0^2)\right]$  and  $z = z_0 \frac{t}{t_0}$ .

These equations express how the vorticity is diluted during the process, with a dominant behavior in  $1/t$ .

---

If the evolution of the vorticity during the expansion is understandable, this is very unclear what will be generated as vorticity at initial stages of the collision: 2 examples below...

